NIH National Institute on Aging

Sample Application for Research Training and Career Development Funding

Through the F99/K00 Transition to Aging Research for Predoctoral Students Awards, NIA supports exceptional graduate students in finishing their dissertation research on any topic and continuing to a postdoctoral position conducting research on aging. Each award has two phases: the F99 phase supporting predoctoral training and the K00 phase supporting postdoctoral training.

Please note that these applications do not reflect the new Fellowship review criteria, which are effective January 25, 2025, as well as other updates to application instructions. Be sure to refer to the latest Fellowship Instructions and the current Notice of Funding Opportunity when preparing an F99/K00 application.

Find more NIA sample applications and information about training and career development funding:
https://www.nia.nih.gov/research/training/sample-applications

SUMMARY STATEMENT
(Privileged Communication)

PROGRAM CONTACT:
Dr Audie Atienza
[redacted]

Release Date: **04/09/2020**
Revised Date:

Application Number: **1 F99 AG068492-01**

YU,KEXIN
University of Southern California
[redacted]

Review Group:	**ZAG1 ZIJ-1 (M2)** **National Institute on Aging Special Emphasis Panel** **Transition to Aging Research Award for Predoctoral Students Review**
Meeting Date:	**03/20/2020**
Council:	**MAY 2020** ***PCC:*** **2BBCHAA**
Requested Start:	**07/01/2020**

Project Title:	**Examining the Longitudinal Influence of the Physical and Social Environments on Social Isolation and Cognitive Health: contextualizing the role of technology**
	***Requested*: null**
Sponsor:	
Department:	**Social Work**
Organization:	**UNIVERSITY OF SOUTHERN CALIFORNIA**
City, State:	**LOS ANGELES CALIFORNIA**
SRG Action:	**++**
Next Steps:	**Visit https://grants.nih.gov/grants/next_steps.htm**
Human Subjects:	**30-Human subjects involved - Certified, no SRG concerns**
Animal Subjects:	**10-No live vertebrate animals involved for competing appl.**
Gender:	**1A-Both genders, scientifically acceptable**
Minority:	**1A-Minorities and non-minorities, scientifically acceptable**
Age:	**3A-No children included, scientifically acceptable**

++NOTE TO APPLICANT: Members of the Scientific Review Group (SRG) were asked to identify those applications with the highest scientific merit, generally the top half. Written comments, criterion scores, and preliminary impact scores were submitted by the assigned reviewers prior to the SRG meeting. At the meeting, the more meritorious applications were discussed and given final impact scores; by concurrence of the full SRG, the remaining applications, including this application, were not discussed or scored. The reviewers' comments (largely unedited by NIH staff) and criterion scores for this application are provided below. Because applications deemed by the SRG to have the highest scientific merit generally are considered for funding first, it is highly unlikely that an application with an ND recommendation will be funded. Each applicant should read the written critiques carefully and, if there are questions about the review or future options for the project, discuss them with the Program Contact listed above.

DESCRIPTION (provided by applicant): Social wellbeing and cognitive health are two pillars of successful aging, and they are interrelated. Three decades of research have consistently shown that social isolation is associated with declines in cognitive functioning. Although existing literature has explored the individual-level risk and protective factors of social isolation and cognitive health, less attention has been paid to how the physical and social characteristics of the living environment affect the experience of social isolation and cognitive health. The Ecological Theory of Aging (ETA) posits environmental factors have significant influences on health and social wellbeing. Negative physical characteristics of the environment were hypothesized to related to increased social isolation and cognitive decline longitudinally. Perceptions of social environments are often affected by the physical environment. Hence, it is hypothesized that the social environment mediated the longitudinal association between the physical environment and social isolation as well as cognitive decline. An increasing amount of research investigated the association between information and communication technology (ICT) use and social isolation, yet whether ICT use would interact with the effect of environmental factors on the risk of social isolation has not yet been empirically tested. Besides, the influence of ICT use on cognitive health has not been well understood. Promoting social interaction using technology-based approaches has the potential to alleviate social isolation among older people and prevent cognitive decline at the same time. The proposed dissertation work (F99) adopts a mixed-methods approach. National Health & Aging Trend Study (NHATS) waves 5 to 8 data will be used to investigate the hypothesized relationships between physical and social environment, ICT use, social isolation, and cognitive decline with latent growth curve modeling. To explain how and why social and physical aspects of the living environment affect social wellbeing and health outcomes of its older residents, focus groups will be conducted to elicit the older adults' perception of the environmental influence on social wellbeing. In the K00 phase, the candidate will use the longitudinal data from the ongoing randomized controlled trial "Internet-based Conversational Engagement Clinical Trial (I-CONECT)" to investigate the effectiveness of an ICT-enhanced conversational intervention on social isolation and its effect on cognitive health among older adults with normal cognition or mild cognitive impairment (MCI). The proposed research will produce knowledge on the influence of environmental factors on social isolation and cognitive health. Knowledge built in F99 and K00 stages will be synthesized to inform the development of a technology-facilitated intervention program for social isolation and cognitive decline from the perspective of ETA.

PUBLIC HEALTH RELEVANCE: Social isolation and cognitive health are interrelated critical outcomes in the aging process, yet their association with the physical and social living environment has not been extensively researched. Promoting social interaction using technology-based approaches has the potential to alleviate social isolation and prevent cognitive decline. The proposed work will produce longitudinal empirical evidence on the influence of environmental factors on social isolation and cognitive health, examining the effectiveness of an information and communication technology-enhanced conversational intervention for social isolation reduction and delay of onset of dementia, and eventually, synthesizing the findings to inform the development of a technology-facilitated intervention program for social isolation and cognitive decline using an ecological approach.

CRITIQUE 1:

Fellowship Applicant: 3
Sponsors, Collaborators, and Consultants: 4
Research Training Plan: 6
Training Potential: 4
Institutional Environment & Commitment to Training: 2

Overall Impact:
This application comes from a PhD student in social work interested in learning to develop information and communication technology (ICT) for older adults to prevent cognitive decline and ADRD. The applicant shows evidence of trying to build a scholarly record, with a couple published co-authored papers and others under review and in preparation. There is also a team with some wide-ranging background spanning social work, demography, anthropology, and engineering. The overall research ideas involve many pieces from community trials to technology to qualitative work to imaging biomarkers. While breadth and integration are admirable goals, many pieces seem somewhat underdeveloped, as reflected in the training plan. Some mentoring and training experience in the clinical world of ADRD (i.e., memory care clinics, neurologists) would be useful given the emphasis on interventions, and specialized mentorship and training is needed for both imaging and advanced statistics in this area since these fields are highly technical and unlikely to be easily acquired skills through limited and/or self-directed approaches.

1. Fellowship Applicant:
Strengths
- The applicant is a social work PhD interested in using information and communication technology to reduce social isolation in older adults, an important area of work.
- There is some background in research assistantships.
- The applicant is learning to utilize common aging data to study these concepts, demonstrating an interest in empirical work.

Weaknesses
- Early signs of productivity are ambiguous with two co-authored publications and three in various stages of review, but many others “in preparation”.

2. Sponsors, Collaborators, and Consultants:
Strengths
- A diverse disciplinary group, including aging, engineering, and anthropology and demography.
- Dr. Dodge has done a considerable amount of work from the population angle on dementia and cognitive aging.

Weaknesses
- The first training goal involves “prevention, intervention and policy” for ADRD, yet there are no geriatric psychiatrists, neurologists, or neuropsychologists or others working clinically with these individuals on the team. Given the centers and entities the applicant lists, such individuals would seem to be available.
- A mentor with expertise in neuroimaging is needed, given the prominence of this in the training aims.
- A biostatistician specializing in imaging data is needed.

3. Research Training Plan:
Strengths
- A systematic training plan is in place.
- A range of activities is planned, including some at well-respected centers.

Weaknesses
- Some more clinical experience with ADRD would be beneficial, so the candidate can understand how screening, intervention, and health system use currently occurs with this set of conditions; neurology is mentioned as an area of training, but there are no neurologists.
- Imaging statistics is a highly specialized field. It is unlikely that anyone without a Master’s in statistics could “self-learn” adequately. More generally, R is a useful tool, but it is a programming language and not synonymous with a true understanding of statistics. Formal quantitative methods coursework in some of the methods proposed here would be desirable.

- The major goal of this mechanism is to produce independent scientists. There is very little time allotted for grant writing, relative to what it actually takes.
- At this stage in the individual's career, and for the purposes of this award, teaching and mentoring should really take a back seat to pursuing novel coursework and training.
- It is a bit unclear what the candidate means in a future goal of obtaining "another career development award"; generally, one wants to do a K, then move on to Rs, preferably an R01. Mid-career K's would be much further down the road.

4. Training Potential:
Strengths
- The candidate appears motivated to pursue this area of work.
- The topic area and mixture of new training is reasonable and would yield a future scholar with a unique mixture of skills.

Weaknesses
- Many aspects of the training plan and/or mentorship team seem not to be fully developed.

5. Institutional Environment & Commitment to Training:
Strengths
- Environment is adequate.

Weaknesses
- Not provided.

Protections for Human Subjects:
Acceptable Risks and Adequate Protections.

Inclusion Plans:
- Sex/Gender: Distribution justified scientifically.
- Race/Ethnicity: Distribution justified scientifically.
- Inclusion/Exclusion Based on Age: Distribution justified scientifically.

Training in the Responsible Conduct of Research:
Acceptable.

Budget and Period of Support:
Recommend as Requested.

CRITIQUE 2:

Fellowship Applicant: 1
Sponsors, Collaborators, and Consultants: 1
Research Training Plan: 1
Training Potential: 4
Institutional Environment & Commitment to Training: 1

Overall Impact:
Based at the University of Southern California, the applicant will examine the association of social isolation and cognition, exploring the moderating influence of information and communication technology. The mixed methods approach has several aspects. The National Health & Aging Trend Study (NHATS) study will allow the investigation of longitudinal associations between physical and social environment, communication technology use, social isolation, and cognitive decline, using state-of-the-art longitudinal methods. This will be augmented with focus groups that probe the effects of the

built environment on wellbeing. The postdoc pivots to an ongoing RCT that is investigating a communication technology-based intervention on cognition in elders with and without MCI. This is a well-regarded candidate with strong academic performance and supportive mentors and referees. The mentoring team is excellent, and multi-institutional (raising some concern about the frequency of contact with at-distant mentors who are essential to the success of this program). The research is quantitatively strong and theoretically grounded, but additional integration of the focus group piece and consideration of the risks/challenges of the intervention postdoctoral study could have been more fully described. The trainee has had many excellent research experiences, but publication productivity has been low.

1. Fellowship Applicant:
Strengths
- The candidate describes rich international experiences and a strong commitment to the science of social work, which has been well recognized with awards.
- Academic performance in relevant coursework has been strong, and the letters are uniformly positive and supportive.

Weaknesses
- None noted.

2. Sponsors, Collaborators, and Consultants:
Strengths
- Drs. Chi and Wu bring experience and productivity in the communications strategies and technology that undergird the application.
- Dr. Palinkas supports the mixed methods/focus group components.
- Dr. Dodge is a recognized expert in the statistical approach described.

Weaknesses
- Given Dr. Dodge's vital role, her monthly conference calls with the participant may need to be augmented in frequency and via other communications strategies.

3. Research Training Plan:
Strengths
- The training plan is superbly detailed; every training activity, timing and frequency are all clearly outlined.
- Didactic and practical experiences seem appropriate.
- The proposed research is theoretically grounded, and a conceptual model is provided.

Weaknesses
- In general, while the focus group component is innovative and the methodology is clearly described, the rationale for adding this piece to an otherwise highly quantitative program of study is less well justified.
- The postdoctoral intervention study is likely to bring certain challenges (e.g., selective loss of participants, protocol deviations), but these are not considered, suggesting that more training in clinical trial methodology might be warranted.

4. Training Potential:
Strengths
- Letters of recommendation document strong clinical and scientific skills, and high enthusiasm for the candidate.

Weaknesses
- Numerous manuscripts are in preparation and under review, but there has been no publication so far.

5. Institutional Environment & Commitment to Training:

Strengths

- The resources, facilities, environment and mentorship are all strong and appropriate for the training.

Weaknesses

- There does not seem to be a commitment to providing specific resources tailored to this fellowship, but the general training environment is excellent.

Protections for Human Subjects:
Acceptable Risks and Adequate Protections.

- All major elements of NIH research protections are thoroughly addressed.

Inclusion Plans:

- Sex/Gender: Distribution justified scientifically.
- Race/Ethnicity: Distribution justified scientifically.
- Inclusion/Exclusion Based on Age: Distribution justified scientifically.
- The study is focused only on older adults, so age exclusions are justified.

Training in the Responsible Conduct of Research:
Acceptable.
Format:

- Four different initiatives are mentioned, and format is discussed in each.

Subject Matter:

- Subject matter is reviewed for all four initiatives.

Faculty Participation:

- Faculty participation is addressed in all four initiatives.

Duration:

- Duration is addressed in all four initiatives.

Frequency:

- Frequency is addressed in all four initiatives.

Budget and Period of Support:
Recommend as Requested.

CRITIQUE 3:

Fellowship Applicant: 3
Sponsors, Collaborators, and Consultants: 1
Research Training Plan: 5
Training Potential: 3
Institutional Environment & Commitment to Training: 1

Overall Impact:
The research will examine the effects of information and communication technology (ICT) use on cognitive functioning. Promoting social interaction using technology-based approaches can potentially alleviate the social isolation among older people and prevent cognitive decline at the same time. The predoctoral work will focus on the longitudinal impact of the environments of living on social isolation and cognitive health with four waves of data from the National Health & Aging Trend Study (NHATS). The postdoctoral work will use longitudinal data from the randomized controlled trial "Internet-based Conversational Engagement Clinical Trial (I-CONECT)" to investigate the effectiveness of an ICT-enhanced conversational intervention on social isolation and its effect on cognitive health among older adults with normal cognition or mild cognitive impairment (MCI). The applicant has reasonable potential

for a productive career. The training plan is consistent with the stated goals. The sponsors and environment provide the necessary resources to support strong predoctoral and postdoctoral programs.

1. Fellowship Applicant:
Strengths
- Graduate student in social work at USC.
- Her grades are good.

Weaknesses
- Most of the publications listed are in preparation. She has just one publication.

2. Sponsors, Collaborators, and Consultants:
Strengths
- All collaborators are strong with excellent qualifications.
- Dissertation is at USC with Drs. Chi, Wu and Palinkas.
- Postdoctoral work is at Oregon Science Institute with Dr. Dodge.
- The sponsors have adequate research funding.

Weaknesses
- None noted.

3. Research Training Plan:
Strengths
- The training plan is clear and well planned.
- The dissertation examines the longitudinal impact of the environments of living on social isolation and cognitive health with four waves of data from the National Health & Aging Trend Study (NHATS). The subjective perception of the environmental influence on social well-being will be elicited using focus groups. It explores the possible designs of a community-based, technology-facilitated approach that could sustainably enhance the social well-being and cognitive health of isolated older adults.
- In the K00 phase, longitudinal data from the ongoing randomized controlled trial “Internet-based Conversational Engagement Clinical Trial (I-CONECT)” will be used to investigate the mechanism of conversational engagement on cognitive functioning. The I-CONECT trial is currently collecting digital biomarker and neuroimaging data, including medication adherence, speech and language characteristics, and region-specific structural and functional pre-post changes in the brain. She will examine why and how social interaction affects the cognitive functioning of older adults as well as how technology can be used effectively to alleviate isolation and improve cognitive functions.
- The F99 phase will examine the effect of physical and social environments on social isolation and cognitive health using a mixed-methods approach. ICT use is hypothesized to moderate the influence of physical and social environments.
- The K00 phase will examine the effectiveness of an ICT-enhanced conversational intervention on social isolation and its effect on cognitive health among older adults with normal cognition or mild cognitive impairment (MCI). Explores the potential to co-design a technology-facilitated community-based intervention from an ecological theory of aging perspective for social isolation and cognitive decline with older adult participants.

Weaknesses
- The study is qualitative based on a focus group.

4. Training Potential:
Strengths
- The proposed research project and training plan will provide the applicant with the necessary experiences to obtain the appropriate skills.

Weaknesses

- Not provided.

5. Institutional Environment & Commitment to Training:
Strengths
- USC is committed to supporting her research.
- Oregon Health Sciences Institute will support the postdoctoral work.
- Facilities are excellent.

Weaknesses
- None noted.

Protections for Human Subjects:
Acceptable Risks and Adequate Protections.

Inclusion Plans:
- Sex/Gender: Distribution justified scientifically.
- Race/Ethnicity: Distribution justified scientifically.
- Inclusion/Exclusion Based on Age: Distribution justified scientifically.

Training in the Responsible Conduct of Research:
Acceptable.

Budget and Period of Support:
Recommend as Requested.

CRITIQUE 4:

Overall Impact:
The applicant has an impressively long interest and research experience in application of web-based tools for the enhancement of cognitive function in older adults, beginning with undergraduate studies at Zhejiang University and continuing through MSW and now PhD studies at USC. She has two publications (others in preparation of under review), strong academic performance and enthusiastic letters of support. She has assembled a diverse and supportive mentoring team, each well published. Dr. Dodge (OHSU) is particularly noteworthy and is exceptionally well-funded, mostly through NIA behavioral intervention studies. That Dr. Dodge and OHSU have committed to be the K00 mentor and training site is a strength of the application. F99 training goals are laudable; however, structured learning is modest. As an example, mentored training in statistics (rather than a self-directed course in R programming) would seem to have been desirable. The research plan has two components: First, examining the effect of information and communication technology (ICT) on social isolation and cognitive health using data from the National Health & Aging Trend Study (NHATS). The second aspect of F99 training leads directly into the K00 aim, although it does not appear that this phase will result in publication(s) during the doctoral training period. This initial work will focus on qualitative and quantitative analysis of focus group data from I-CONECT subjects to form a foundation for developing an intervention program to be tested in the postdoctoral training phase. This K00 phase will conduct co-design sessions with isolated older adults who are at risk of cognitive decline. While use and training in neuroimaging are described in the K00 training plan, it is not clear that this is integrated into the K00 research plan. As seem from the above, the training potential is good, although several weaknesses and ambiguities in it somewhat reduce enthusiasm. The institutional environmentF at both USC and OHSU is excellent. In summary, the applicant is highly committed to a career in research to enhance the cognitive function in older adults. Strengths of the application include the mentoring team, the continuity and focus of both F99 and K00 work on cognitive aging, and the strong institutional

environments and support. The training plan and potential, while satisfactory, are uneven in several respects. The latter reduces the otherwise outstanding impact to a more moderated excellent level.

Footnotes for 1 F99 AG068492-01; PI Name: Yu, Kexin

NIH has modified its policy regarding the receipt of resubmissions (amended applications).See Guide Notice NOT-OD-18-197 at https://grants.nih.gov/grants/guide/notice-files/NOT-OD-18-197.html. The impact/priority score is calculated after discussion of an application by averaging the overall scores (1-9) given by all voting reviewers on the committee and multiplying by 10. The criterion scores are submitted prior to the meeting by the individual reviewers assigned to an application, and are not discussed specifically at the review meeting or calculated into the overall impact score. Some applications also receive a percentile ranking. For details on the review process, see http://grants.nih.gov/grants/peer_review_process.htm#scoring.